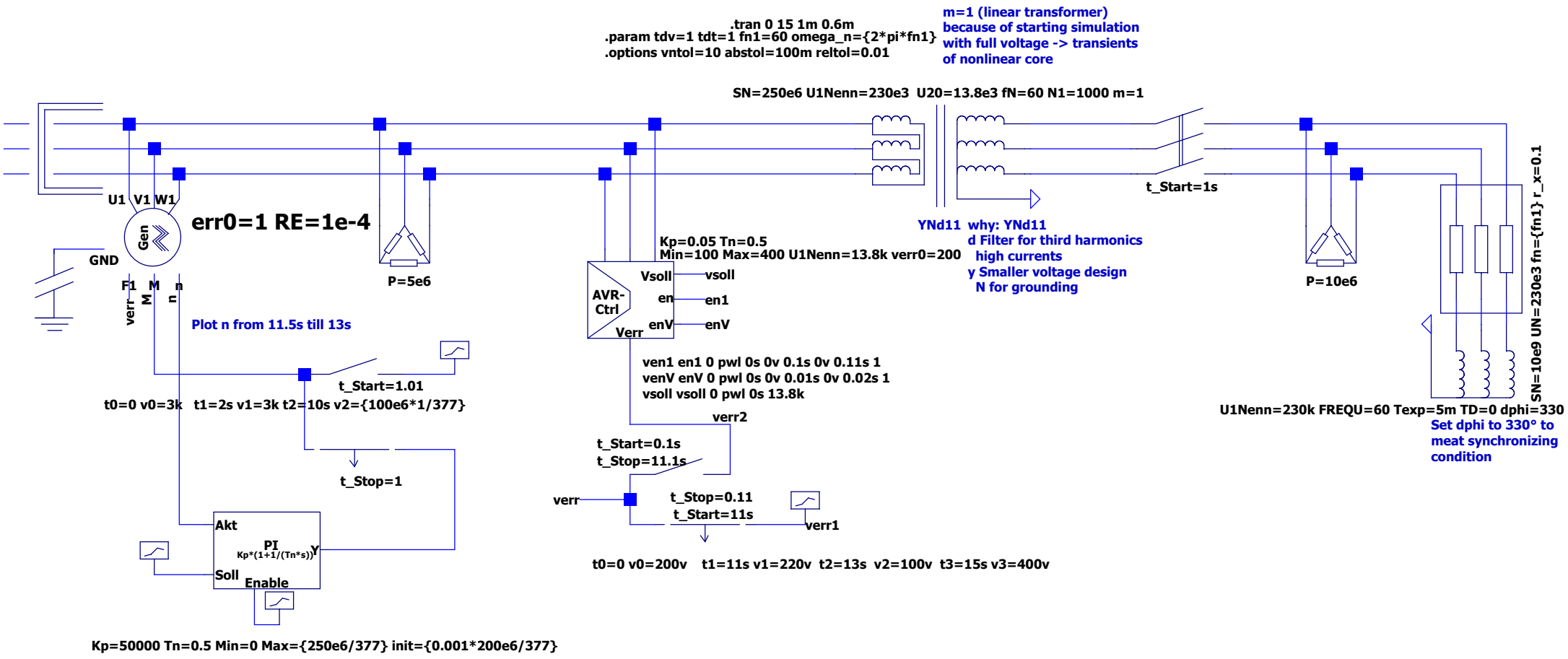
.tran 0 15 1m 0.6m
.param tdv=1 tdt=1 fn1=60 omega_n={2*pi*fn1}
.options vntol=10 abstol=100m reltol=0.01
m=1 (linear transformer)
because of starting simulation
with full voltage -> transients
of nonlinear core
SN=250e6 U1Nenn=230e3 U20=13.8e3 fN=60 N1=1000 m=1
U1 V1 W1
Gen
GND
err0=1 RE=1e-4
F1 M n
verr
M
n
P=5e6
Kp=0.05 Tn=0.5
Min=100 Max=400 U1Nenn=13.8k verr0=200
AVR-
Ctrl
Vsoll
en
enV
Verr
vsoll
en1
enV
YNd11 why: YNd11
d Filter for third harmonics
high currents
y Smaller voltage design
N for grounding
t_Start=1s
P=10e6
SN=10e9 UN=230e3 fn={fn1} r_x=0.1
U1Nenn=230k FREQU=60 Texp=5m TD=0 dphi=330
Set dphi to 330° to
meat synchronizing
condition
Plot n from 11.5s till 13s
t_Start=1.01
t0=0 v0=3k t1=2s v1=3k t2=10s v2={100e6*1/377}
t_Stop=1
Akt
PI
Kp*(1+1/(Tn*s))
Y
Soll
Enable
Kp=50000 Tn=0.5 Min=0 Max={250e6/377} init={0.001*200e6/377}
ven1 en1 0 pwl 0s 0v 0.1s 0v 0.11s 1
venV enV 0 pwl 0s 0v 0.01s 0v 0.02s 1
vsoll vsoll 0 pwl 0s 13.8k
verr2
t_Start=0.1s
t_Stop=11.1s
verr
t_Stop=0.11
t_Start=11s
verr1
t0=0 v0=200v t1=11s v1=220v t2=13s v2=100v t3=15s v3=400v